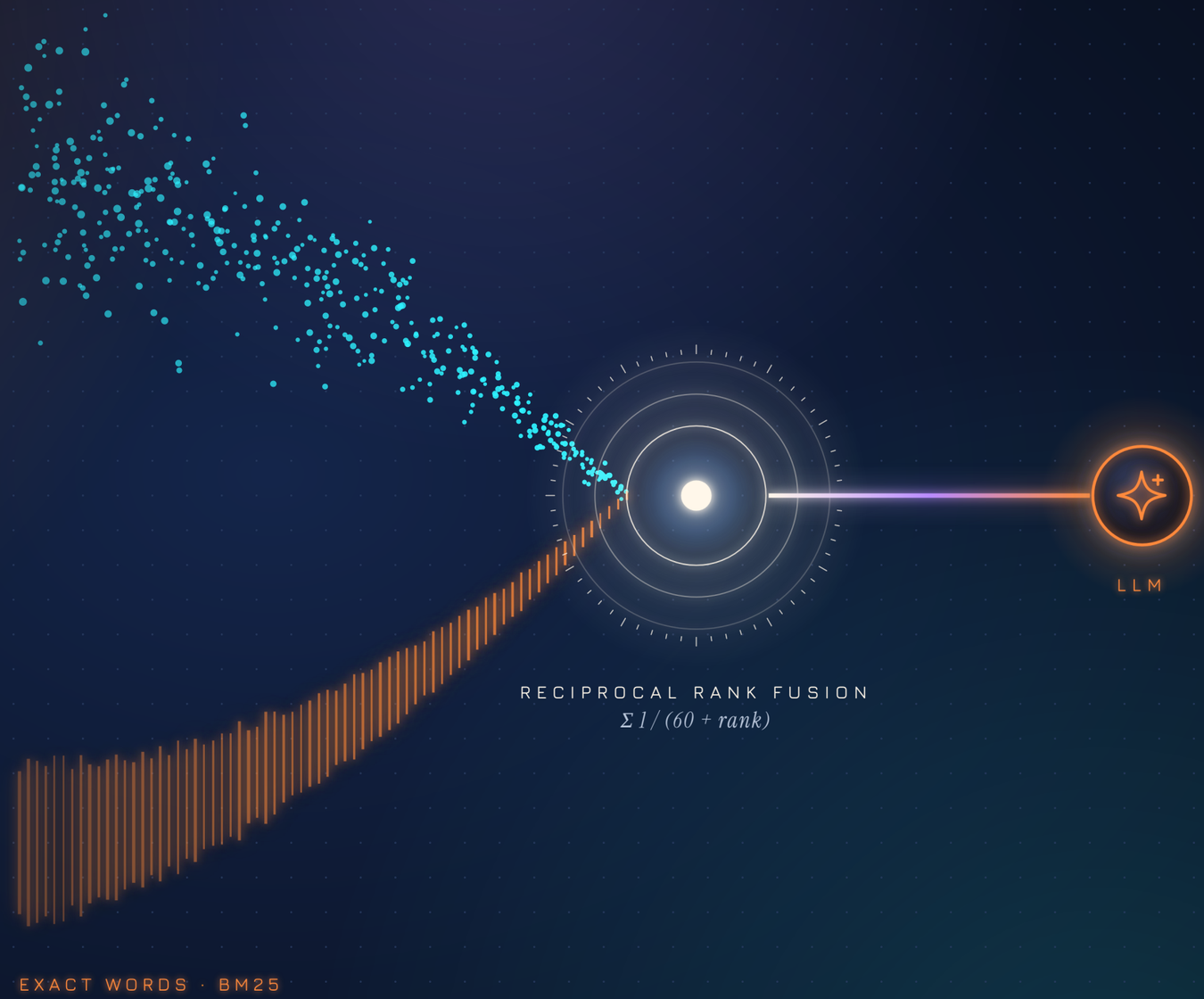
FusionRAG

HYBRID RETRIEVAL · AUGMENTED GENERATION · CLAUDE

Cited answers *from our own documents.*

II Architecture & packet flow 02
III Retrieval methods 03
IV Frontend & backend 04
V Build & deployment 05
VI Operations 06



2 CONTAINERS

3 RETRIEVAL MODES

7 NETWORK HOPS

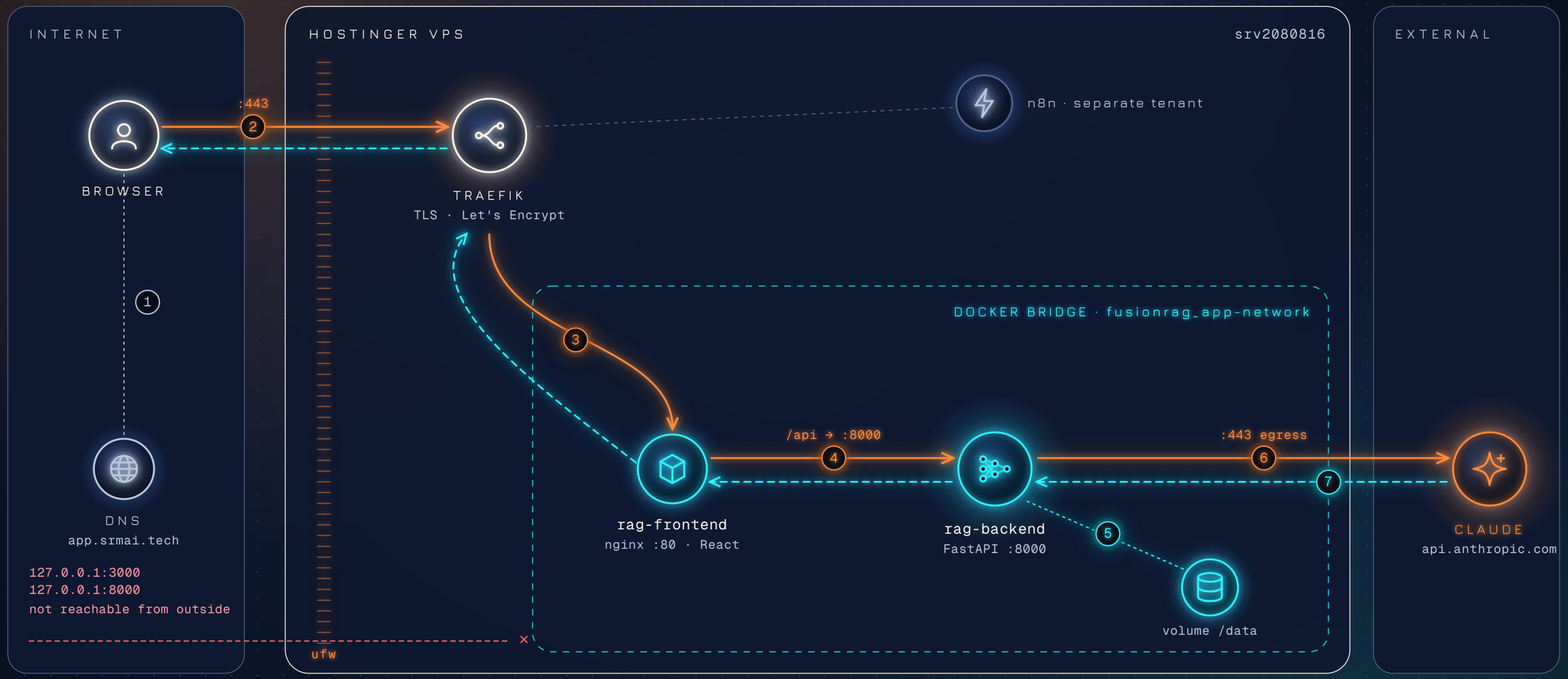
24 BUILD STEPS

Suresh Ramalingam
SENIOR IT INFRASTRUCTURE SME · RRD GDC CHENNAI
OCTOBER 2026

Architecture & Packet Flow

ONE QUESTION, SEVEN HOPS, BROWSER TO CLAUDE AND BACK

IPv4 187.126.126.237 · UBUNTU 24.04
FUSIONRAG · SIGNAL CARTOGRAPHY



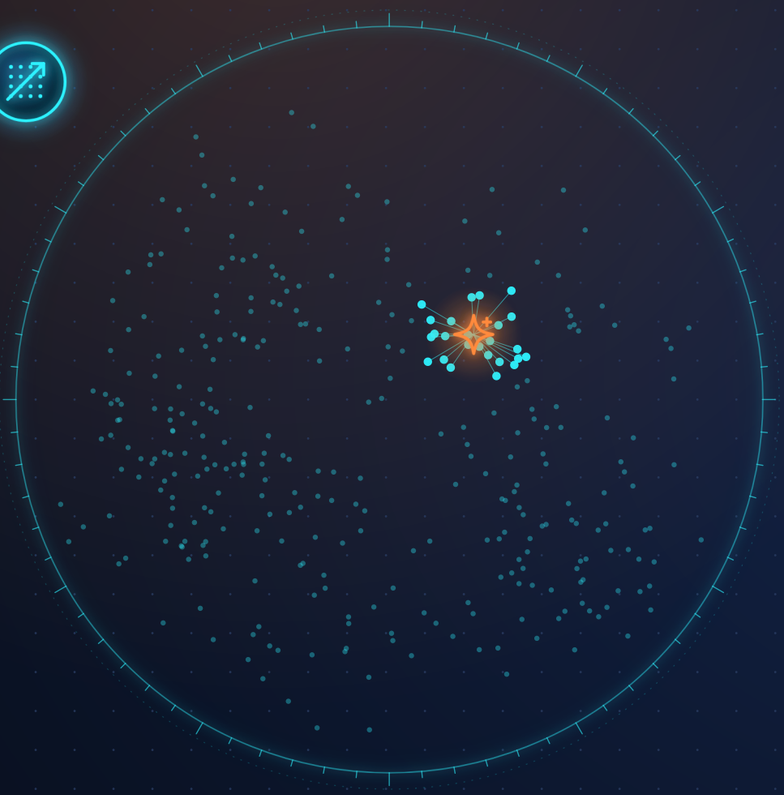
01 RESOLVE	02 ENTER	03 ROUTE	04 PROXY	05 SEARCH	06 GENERATE	07 STREAM
DNS A record	HTTPS :443	Host rule → :80	/api strip	hybrid index	Claude :443	SSE tokens

REQUEST · ANSWER STREAM · PERSIST · BLOCKED

Only 80 and 443 face the world.

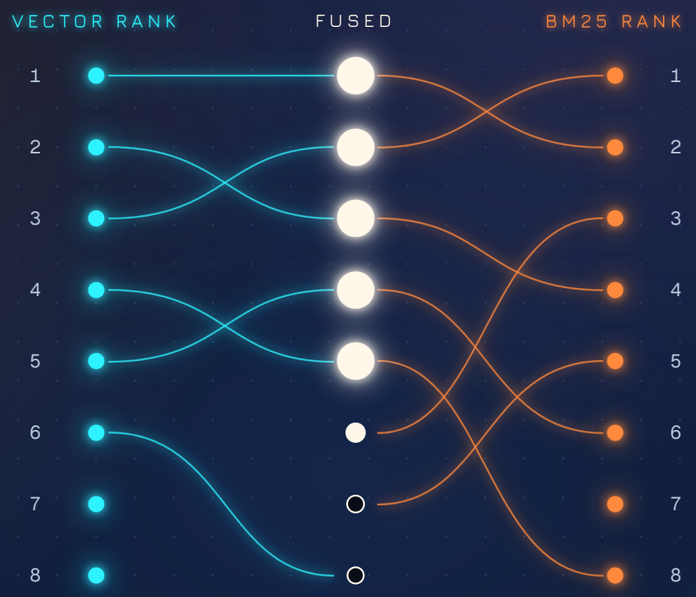
Retrieval Methods

TWO WAYS OF READING, ONE ORDER OF TRUST



VECTOR · MEANING

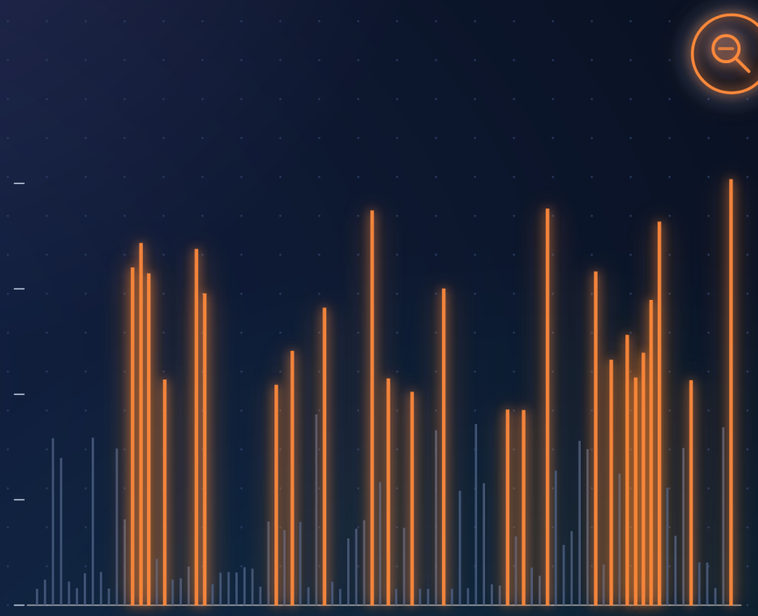
bge-small-en-v1.5 · cosine · top 25



● top 6 → Claude

$$\Sigma\, 1/(60 + rank)$$

agreement rises · scales never compared



BM25 · EXACT WORDS

\w+ tokens · stop words out · top 25 with score > 0

"19:1131" → 19 · 1131

HYBRID — both lists

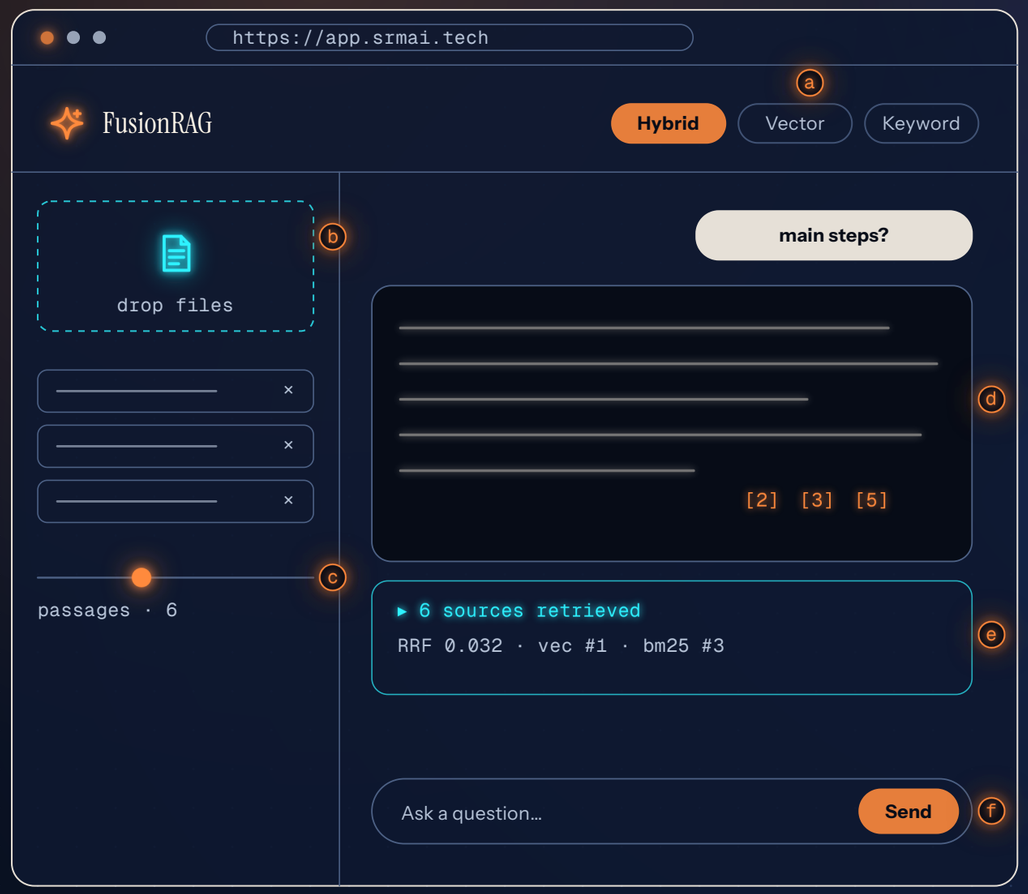
VECTOR — meaning only

KEYWORD — words only

6 passages → ‹source› prompt → Claude → cited stream

Frontend & Backend

WHAT THE USER TOUCHES · WHAT THE SERVER REMEMBERS







INTERFACE

a mode switch · b upload ≤ 25 MB · c top-k 2-12 · d live markdown · e rank evidence · f stop
/api relative · fetch + ReadableStream · nginx SPA fallback · buffering off

ENDPOINTS

GET	/health	DELETE	/documents/{id}
GET	/config	POST	/search
GET	/documents	POST	/chat
POST	/documents		

● INPUT ● TRANSFORM ● MEMORY

Embedding outside the lock: chat never waits for an upload.

Build & Deployment

TWENTY-FOUR STEPS · SEVEN PHASES · EACH ONE VERIFIED



LAPTOP
docker compose up --build
steps 1-15

GITHUB
commit · push · private repo
step 16

VPS
setup-vps.sh · .env × 2
steps 17-20

LIVE
https://app.srmai.tech
steps 21-24

CHECKS

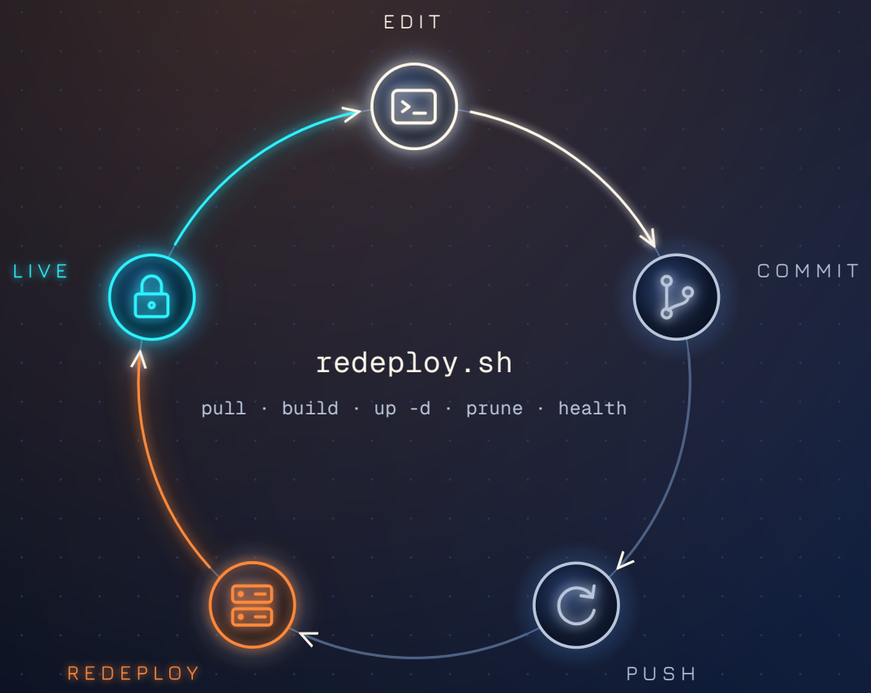
- engine up
- /api/health → ok
- 5 traefik labels
- valid certificate
- 4/4 up after reboot

```
$ docker compose up --build -d
```

Rebuilt from a clean clone. It came back on its own.

Operations

RELEASE · OBSERVE · RECOVER · PROTECT

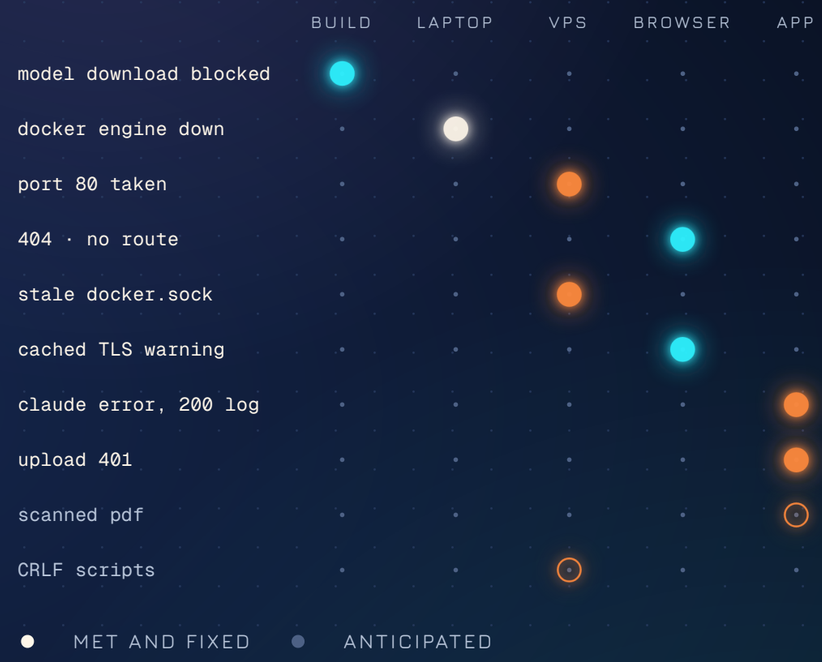




THE INDEX PERSISTS

- rebuild
- redeploy
- reboot
- down -v

TROUBLESHOOTING ATLAS



OPEN SECURITY ACTIONS

- HIGH — admin token
- HIGH — n8n port 32768
- MED — chat rate limit
- MED — env backup file
- LOW — ssh keys

OBSERVE

```
$ docker compose logs -f backend
$ logs backend | grep ERROR|Traceback
$ docker logs traefik-traefik-1 --since 10m
$ curl 127.0.0.1:3000/api/health
```

/chat logs 200 even when Claude fails: read the ERROR line.